

I. Connecting components on a board

V3S[1] PG4/SDC1_DAT2	- EP1C6 [107]	eMMC_U32[A5] DAT2	- EP1C6 [34]
V3S[128] PG5/SDC1_DAT3	- EP1C6 [108]	eMMC_U32[B2] DAT3	- EP1C6 [31]
V3S[4] PG1/SDC1_CMD	- EP1C6 [111]	eMMC_U32[M5] CMD	- EP1C6 [35]
V3S[5] PG0/SDC1_SCK	- EP1C6 [112]	eMMC_U32[M6] SCK	- EP1C6 [36]
V3S[3] PG2/SDC1_DAT0	- EP1C6 [110]	eMMC_U32[A3] DAT0	- EP1C6 [32]
V3S[2] PG3/SDC1_DAT1	- EP1C6 [109]	eMMC_U32[A4] DAT1	- EP1C6 [33]

V3S[39] PB0/UART2_TX	- Y4[26]
V3S[40] PB1/UART2_RX	- Y4[28]
V3S[41] PB2/UART2_RTS	- Y4[30]
V3S[42] PB3/UART2_CTS	- Y4[32]
V3S[43] PB4/PWM0	- Y4[34] - R69 - U22_PT4103[04]ENABLE_LCD_V_LED
V3S[44] PB5/PWM1	- Y4[36]
V3S[45] PB6/TWI0_SCL	- NS2009_U16[10] - Y4[38] - FPC06_top_TP5''[06] - FPC06_bottom_TP4.3''[04]
V3S[46] PB7/TWI0_SDA	- NS2009_U16[09] - Y4[40] - FPC06_top_TP5''[05] - FPC06_bottom_TP4.3''[05]
V3S[48] PB8 /UART0_TX/TWI1_SCL	- Y2[03] - Y4[42]
V3S[49] PB9 /UART0_RX/TWI1_SDA	- Y2[01] - Y4[44]
V3S[09] PE21/TWI1_SCL	- - Y4[46]
V3S[08] PE22/TWI1_SDA	- - Y4[48]

V3S[52] PC0/SPI0_MISO	- GD5F1G_U19[02] - X10[02]
V3S[53] PC1/SPI0_SCK	- GD5F1G_U19[06]
V3S[54] PC2/SPI0_CS	- GD5F1G_U19[01]
V3S[55] PC3/SPI0_MOSI	- GD5F1G_U19[05]

STM32F412[42] PA9 **USART1_TXD** - U3_SIT3232[10] -> [07] - X1[02]
STM32F412[43] PA10 **USART1_RXD** - U3_SIT3232[09] <- [08] - X1[01]
STM32F412[51] PC10 **USART3_TXD** - **USART1_RXD** GD32F103[10] /USB-UART-CH2/
STM32F412[25] PC5 **USART3_RXD** - **USART1_TXD** GD32F103[09] /USB-UART-CH2/

STM32F412[54] PD2 SPI1_CS	- W25Q64_U9[01]
STM32F412[55] PB3 SPI1_SCK	- W25Q64_U9[06]
STM32F412[56] PB4 SPI1_MISO	- W25Q64_U9[02]
STM32F412[23] PA7 SPI1_MOSI	- W25Q64_U9[05]

STM32F412[58] PB6 I2C1_SCL	- M24C64_U5[06]
STM32F412[59] PB7 I2C1_SDA	- M24C64_U5[05]

STM32F412[11] PC3 UART2_TR	- SIT3485_U1[2,3]
STM32F412[16] PA2 UART2_TXD	- SIT3485_U1[04]
STM32F412[17] PA3 UART2_RXD	- SIT3485_U1[01]

STM32F412[57] PB5 CAN2_RX	- NCA1042_U2[04]
STM32F412[34] PB13 CAN2_TX	- NCA1042_U2[01]

STM32F412[33] PB12 SPI3_SCK	- EP1C6[128]
STM32F412[52] PC11 SPI3_MISO	- EP1C6[122]
STM32F412[53] PC12 SPI3_MOSI	- EP1C6[121]
STM32F412[41] PA8 SPI3_CS	- EP1C6[17] I.13

EP1C6[119] **RX** - X13[02] <- STM32F412[37]_PC6_ **USART6_TXD** & U3_SIT3232[12] <- [13] - X1[04]
EP1C6[113] **TX** - X13[03] -> STM32F412[38]_PC7_ **USART6_RXD** - U3_SIT3232[11] -> [14] - X1[05]
EP1C6[120] **CK** - STM32F412[39]_PC8_ **USART6_CK**

EP1C6[49] **ADS_RST** - ADS1248_U33[04]
EP1C6[40] **ADS_STRT** - ADS1248_U33[23]
EP1C6[41] **ADS_RDY** - ADS1248_U33[25]
EP1C6[39] **SPI_CS0** - ADS1248_U33[24]
EP1C6[37] **SPI_MISO** - ADS1248_U33[26]
EP1C6[38] **SPI_MOSI** - ADS1248_U33[27] - DAC8830_U27[06] - DAC8830_U30[06]
EP1C6[06] **SPI_SCK** - ADS1248_U33[28] - DAC8830_U27[05] - DAC8830_U30[05]
EP1C6[50] **SPI_CS1** - DAC8830_DACA_U30[04]
EP1C6[51] **SPI_CS2** - DAC8830_DACB_U27[04]

EP1C6[62] **ADCA_CLK** - ADS7884_U34[04]
EP1C6[93] **ADCA_DAT** - ADS7884_U34[05]
EP1C6[61] **ADCA_CS** - ADS7884_U34[06]
EP1C6[72] **ADCB_CLK** - ADS7884_U35[04]
EP1C6[92] **ADCB_DAT** - ADS7884_U35[05]
EP1C6[73] **ADCB_CS** - ADS7884_U35[06]

MT48LC16M16[29] **A4** - EP1C6[07]
MT48LC16M16[30] **A5** - EP1C6[10]
MT48LC16M16[31] **A6** - EP1C6[11]
MT48LC16M16[32] **A7** - EP1C6[12]
MT48LC16M16[23] **A0** - EP1C6[25]
MT48LC16M16[24] **A1** - EP1C6[26]
MT48LC16M16[25] **A2** - EP1C6[27]
MT48LC16M16[26] **A3** - EP1C6[28]
MT48LC16M16[22] **A10** - EP1C6[42]
MT48LC16M16[21] **BA1** - EP1C6[47]
MT48LC16M16[20] **BA0** - EP1C6[48]
MT48LC16M16[18] **RAS** - EP1C6[52]
MT48LC16M16[17] **CAS** - EP1C6[53]
MT48LC16M16[16] **WE** - EP1C6[56]
MT48LC16M16[15] **DQML** - EP1C6[57]
MT48LC16M16[13] **DQ7** - EP1C6[58]
MT48LC16M16[11] **DQ6** - EP1C6[59]
MT48LC16M16[10] **DQ5** - EP1C6[60]
MT48LC16M16[08] **DQ4** - EP1C6[67]
MT48LC16M16[07] **DQ3** - EP1C6[68]
MT48LC16M16[05] **DQ2** - EP1C6[69]
MT48LC16M16[04] **DQ1** - EP1C6[70]
MT48LC16M16[02] **DQ0** - EP1C6[71]
MT48LC16M16[53] **DQ15** - EP1C6[98]
MT48LC16M16[51] **DQ14** - EP1C6[103]
MT48LC16M16[50] **DQ13** - EP1C6[104]
MT48LC16M16[48] **DQ12** - EP1C6[105]
MT48LC16M16[47] **DQ11** - EP1C6[106]
MT48LC16M16[45] **DQ10** - EP1C6[114]
MT48LC16M16[44] **DQ9** - EP1C6[123]
MT48LC16M16[42] **DQ8** - EP1C6[124]
MT48LC16M16[39] **DQMH** - EP1C6[125]
MT48LC16M16[36] **A12** - EP1C6[133]
MT48LC16M16[38] **CLK** - EP1C6[134]
MT48LC16M16[35] **A11** - EP1C6[139]
MT48LC16M16[34] **A9** - EP1C6[140]
MT48LC16M16[33] **A8** - EP1C6[141]

W25Q32_U7[01] - **SPI_CS** - GD32F103_U6 [29]
W25Q32_U7[06] - **SPI_SCK** - GD32F103_U6 [30]
W25Q32_U7[02] - **SPI_MISO** - GD32F103_U6 [31]
W25Q32_U7[05] - **SPI_MOSI** - GD32F103_U6 [32]

EP1C6[77] IO.25 - R117- T3 - LD4_ **LED WHITE**
OSC_25MHz_Q7[03] **CLOCK** - EP1C6[16]

RST_M - STM32F412[07] - GD32F103[04] - V3S[99] - eMMC_U32[K5] - X9[02] - Y2[09]
CLRWDT - STM32F412[39] - EP1C6[120] - T1-RS806_U11[04]

II. Connecting components using jumpers

V3S[48] - **PB8 /UART0_TX** - Y2[03] ----- Y2[04] - **USART0_RXD** - GD32F103[22] /USB-UART-CH1/
V3S[49] - **PB9 /UART0_RX** - Y2[01] ----- Y2[02] - **USART0_TXD** - GD32F103[21] /USB-UART-CH1/

GD32F103[08] - **SWDIOc** - Y3[05] ----- Y3[06] - **SWDIO** - STM32F412[46]
GD32F103[07] - **SWCLKc** - Y3[07] ----- Y3[08] - **SWCLK** - STM32F412[49]

V3S[39] - **PB0/UART2_TX** - Y4[26] ----- Y4[25] - **IO_00** - EP1C6[129]
V3S[40] - **PB1/UART2_RX** - Y4[28] ----- Y4[27] - **IO_01** - EP1C6[130]
V3S[41] - **PB2/UART2_RTS** - Y4[30] ----- Y4[29] - **IO_02** - EP1C6[131]
V3S[42] - **PB3/UART2_CTS** - Y4[32] ----- Y4[31] - **IO_03** - EP1C6[132]
V3S[43] - **PB4/PWM0** - ENA_LED - Y4[34] ----- Y4[33] - **IO_04** - EP1C6[142]
V3S[44] - **PB5/PWM1** - Y4[36] ----- Y4[35] - **IO_05** - EP1C6[143]
V3S[45] - **PB6/TWI0_SCL** - Y4[38] ----- Y4[37] - **IO_06** - EP1C6[144]
V3S[46] - **PB7/TWI0_SDA** - Y4[40] ----- Y4[39] - **IO_07** - EP1C6[01]
V3S[48] - **PB8 /UART0_TX** - Y2[03] - Y4[42] ----- Y4[41] - **IO_08** - EP1C6[02]
V3S[49] - **PB9 /UART0_RX** - Y2[01] - Y4[44] ----- Y4[43] - **IO_09** - EP1C6[03]
V3S[09] - **PE21/TWI1_SCL** - Y4[46] ----- Y4[45] - **IO_10** - EP1C6[04]
V3S[08] - **PE22/TWI1_SDA** - Y4[48] ----- Y4[47] - **IO_11** - EP1C6[05]

RS806[01]-RESET_OUT-X9[01] ----- X9[02] **RST_M** ---- X9[03] - BUTTOM_RST[1]

Set X9[01]---X9[02] - the reset signal RST_M is generated by the watchdog timer and the RESET button

Set X9[02]---X9[03] - the reset signal RST_M is generated only by the RESET button

III. Connecting components for the testing program

- Powering from USB :

X7[01]-----X7[02]

- Powering RTC V3S :

X12[01]-----X12[02]

- Watchdog ON :

X9[01]-----X9[02]

- Connecting the V3S processor's serial port to USB-UART-CH1 of system controller

Y2[01]---Y2[02]

Y2[03]---Y2[04]

- Connecting a system debugger to a microcontroller STM32F412

Y3[05]---Y3[06] - SWDIO

Y3[07]---Y3[08] - SWCLK